

# Chi-Jung LEE

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## RESEARCH INTEREST

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Human-Computer Interaction (HCI), Ubiquitous Computing (UbiComp), Sensing, Wearable, Tangible, E-Textile

## EDUCATION

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### Cornell University

*Ph.D. Student in Information Science*

- Advisor: [Cheng Zhang](#)

Aug. 2022 – Present

Ithaca, New York, USA

### National Taiwan University (NTU)

*Master of Science in Computer Science and Information Engineering*

- Master's Thesis Advisor: [Bing-Yu Chen](#)
- Thesis: HairTouch: Providing Stiffness, Roughness and Surface Height Differences Using Reconfigurable Brush Hairs on a VR Controller
- Graduation GPA: 4.12/4.3

Sep. 2019 – Jun. 2021

Taipei, Taiwan

### National Tsing Hua University (NTHU)

*Bachelor of Science in Computer Science*

*Bachelor of Business Administration in Management and Technology*

- Undergraduate Research Advisor: [Hung-Kuo Chu](#) (Feb. 2018 – Jun. 2019)
- Received **Academic Excellence Award** (honored for top 5% in the department) three times over four years
- Graduation GPA: 4.06/4.3

Sep. 2015 – Jun. 2019

Hsinchu, Taiwan

## PEER-REVIEWED FULL PAPERS

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Kian Mahmoodi, Yudong Xie, Tan Gemicioglu, [Chi-Jung Lee](#), Jiwan Kim, and Cheng Zhang (2025). EchoForce: Continuous Grip Force Estimation from Skin Deformation Using Active Acoustic Sensing on a Wristband  
*ISWC '25: Proceedings of the 2025 ACM International Symposium on Wearable Computers, Espoo, Finland*

[Chi-Jung Lee](#), Jiaxin Li, Tianhong Catherine Yu, Ruidong Zhang, Vipin Gunda, François Guimbretière, and Cheng Zhang (2025). Grab-n-Go: On-the-Go Microgesture Recognition with Objects in Hand  
*IMWUT '25: Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*

Yuan-Chia Chang, Daniel J Rea, [Chi-Jung Lee](#), Takayuki Kanda, Bing-Yu Chen (2025). Designing Authoritative Presence in Social Robots  
*ACM Transactions on Human-Robot Interaction*

Tianhong Catherine Yu, Manru Mary Zhang, Luis Miguel Malenab, [Chi-Jung Lee](#), Jacky Hao Jiang, Ruidong Zhang, François Guimbretière, and Cheng Zhang (2025). SeamFit: Towards Practical Smart Clothing for Automatic Exercise Logging  
*IMWUT '25: Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*

Tianhong Catherine Yu, Guilin Hu, Ruidong Zhang, Hyunchul Lim, Saif Mahmud, [Chi-Jung Lee](#), Ke Li, Devansh Agarwal, Shuyang Nie, Jinseok Oh, François Guimbretière, and Cheng Zhang (2024). Ring-a-Pose: A Ring for Continuous Hand Pose Tracking  
*IMWUT '24: Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*

Tianhong Catherine Yu, Manru Mary Zhang\*, Peter He\*, **Chi-Jung Lee**, Cassidy Cheesman, Saif Mahmud, Ruidong Zhang, François Guimbretière, Cheng Zhang (2024). SeamPose: Repurposing Seams as Capacitive Sensors in a Shirt for Upper-Body Pose Tracking (\*Contributed Equally)

*UIST '24: The 37th Annual ACM Symposium on User Interface Software and Technology, Pittsburgh, PA, USA*

**Chi-Jung Lee\***, Ruidong Zhang\*, Devansh Agarwal, Tianhong Catherine Yu, Vipin Gunda, Oliver Lopez, James Kim, Sicheng Yin, Boao Dong, Ke Li, Mose Sakashita, François Guimbretière, Cheng Zhang (2024). EchoWrist: Continuous Hand Pose Tracking and Hand-Object Interaction Recognition Using Low-Power Active Acoustic Sensing On a Wristband (\*Contributed Equally)

*CHI '24: Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems, Honolulu, HI, USA*

**Chi-Jung Lee**, David Yang, Pin-Sung Ku, Hsin-Liu (Cindy) Kao (2024). SweatSkin: Rapidly Prototyping Sweat-Sensing On-Skin Interface Based on Microfluidics

*IMWUT '24: Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*

**Chi-Jung Lee**, Chi-Huan Chiang, Ling-Chien Yang, Te-Yen Wu, Rong-Hao Liang, Bing-Yu Chen (2022).

NFCStack: Identifiable Physical Building Blocks that Support Concurrent Construction and Frictionless Interaction

*UIST '22: The 35th Annual ACM Symposium on User Interface Software and Technology, Bend, OR, USA*

Huizhong Ye\*, **Chi-Jung Lee\***, Te-Yen Wu\*, Xing-Dong Yang, Bing-Yu Chen, Rong-Hao Liang (2022).

Body-Centric NFC: Body-Centric Interaction with NFC Devices Through Near-Field Enabled Clothing (\*Contributed Equally)

*DIS '22: Designing Interactive Systems Conference, Virtual, Australia*

**Chi-Jung Lee**, Hsin-Ruey Tsai, Bing-Yu Chen (2021). HairTouch: Providing Stiffness, Roughness and Surface Height Differences Using Reconfigurable Brush Hairs on a VR Controller

*CHI '21: Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems, Yokohama, Japan*

Chin-Yuan Lu\*, Han-Wei Hsieh\*, Rong-Hao Liang\*, **Chi-Jung Lee**, Ling-Chien Yang, Mengru Xue, Jr-Ling Guo, Meng-Ju Hsieh, Bing-Yu Chen (2021). Combining Touchscreens with Passive Rich-ID Building Blocks to Support Context Construction in Touchscreen Interactions (\*Contributed Equally)

*CHI '21: Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems, Yokohama, Japan*

## PEER-REVIEWED CONFERENCE ADJUNCT PUBLICATIONS

Jiwan Kim\*, **Chi-Jung Lee\***, Xiaoran Fan, Dong Li, Edison Thomaz, Alanson Sample, Ian Oakley, Cheng Zhang (2025). BeyondSound: The First Workshop on Intelligent Acoustic Sensing on Wearables (\*Contributed Equally)

*UbiComp '25: Companion of the ACM International Joint Conference on Pervasive and Ubiquitous Computing*

Waki Kamino\*, Andrea W Wen-Yi\*, Dhruv Agarwal, Sil Hamilton, Eun Jeong Kang, Jieun Kim, Keigo Kusumegi, Pegah Moradi, Daniel Mwesigwa, Yan Tao, I-Ting Tsai, Ethan Yang, Shengqi Zhu, Shu-Jung Han, **Chi-Jung Lee**, Michael Joseph Sack, Tianhong Catherine Yu, Weslie Khoo, Andy Elliot Ricci, Yoyo Tsung-Yu Hou, Boyoung Kim, Selma abanovic, David J Crandall, Karen Levy, Malte F Jung (2025). Million Eyes on the Robot Umps: The Case for Studying Sports in HRI Through Baseball (\*Contributed Equally)

*HRI '25: Proceedings of the 2025 ACM/IEEE International Conference on Human-Robot Interaction*

🏆 **Chi-Jung Lee**, Hung-Kuo Chu (2018). Dual-MR: Interaction with Mixed Reality Using Smartphones

*VRST '18: Proceedings of the 24th ACM Symposium on Virtual Reality Software and Technology, Tokyo, Japan*  
(Best Poster Award)

## HONOR & AWARD

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### **ETH's Student Summer Research Fellowship**

Jul. 2020 – Aug. 2020

*One of the 15-20 accepted applicants selected out of 2880 applicants*

ETH Zurich, Zurich, Switzerland

- The fellowship provided by the Department of Computer Science at Eidgenössische Technische Hochschule Zürich (ETH Zurich) for select students to pursue a research project as part of a research group in the department

## RESEARCH EXPERIENCE

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### **SciFi Lab, Cornell University**

May. 2023 – Present

*Ph.D. Student*

Ithaca, New York, USA

- Advisor: [Cheng Zhang](#)
- Developing smart sensing wearables

### **Hybrid Body Lab, Cornell University**

Aug. 2022 – May. 2023

*Ph.D. Student*

Ithaca, New York, USA

- Advisor: [Cindy Hsin-Liu Kao](#)
- Developed on-skin interfaces for sweat sensing (Published in IMWUT, Vol. 7, No. 4)

### **CREST, The University of Tokyo**

Apr. 2022 – Jul. 2022

*Research Assistant*

Tokyo, Japan

- Advisor: [Takeo Igarashi](#)
- Developed a design tool that facilitates the design and fabrication of deformable and stretchable circuits on hand-knitted fabrics

### **Interactive Graphics Laboratory, NTU**

Jul. 2021 – Apr. 2022

*Research Assistant*

Taipei, Taiwan

- Advisor: [Bing-Yu Chen](#)
- Developed a scalable coil extender design on skin or garments that enables body-centric interactions for a commodity near-field communication (NFC) device (Published in DIS '22)
- Developed a portable rich-ID building block system based on NFC technology (Published in UIST '22)

### **Interactive Graphics Laboratory, NTU**

Sep. 2019 – Jun. 2021

*Master's Student*

Taipei, Taiwan

- Advisor: [Bing-Yu Chen](#)
- Developed a pin-based handheld device providing tactile feedback using cosmetic brush hairs in virtual reality (Published in CHI '21)
- Developed two building block systems based on the ultra-high frequency (UHF) radio-frequency identification (RFID) sensing, rear-projection, and capacitive touch sensing in collaboration with Prof. [Rong-Hao Liang](#) from Eindhoven University of Technology (Published in CHI '21)
- Built batteryless, wireless, and modularized 2D touch sensor pads based on the UHF RFID technology

### **HCI Lab, NTU**

Feb. 2020 – Aug. 2020

*Project Collaborator*

Taipei, Taiwan

- Advisor: [Mike Y. Chen](#)
- Developed a VR haptic device improving the comfort and the overall first-person viewing (FPV) experience

### **Computer Graphics & Vision Lab, NTHU**

Feb. 2018 – Jun. 2019

*Undergraduate Research Assistant*

Hsinchu, Taiwan

- Advisor: [Hung-Kuo Chu](#)
- Developed a mixed reality interaction system enabling co-located collaboration of a head-mounted device and handheld smartphone (Poster Presentation at VRST '18)

## **CoAI, Tsinghua University**

### *Visiting Student*

Jul. 2017 – Aug. 2017

Beijing, China

- Advisor: Minlie Huang
- Analyzed the algorithms and applications of four off-the-shelf chatbots
- Collected and labeled 5000+ dialogues for the development of a chatbot

## **TEACHING ASSISTANCE**

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### **Introduction to Rapid Prototyping and Physical Computing**

*INFO 4320 / INFO 5321, Prof. Cheng Zhang*

Aug. 2025 – Dec 2025

Cornell University, Ithaca, NY, USA

### **Ubiquitous Computing**

*INFO 4120 / INFO 6120, Prof. Cheng Zhang*

Jan. 2025 – May 2025

Cornell University, Ithaca, NY, USA

### **Introduction to Rapid Prototyping and Physical Computing**

*INFO 4320 / INFO 5321, Prof. Cheng Zhang*

Aug. 2024 – Dec 2024

Cornell University, Ithaca, NY, USA

### **Introduction to Rapid Prototyping and Physical Computing**

*INFO 4320 / INFO 5321, Prof. Cheng Zhang*

Jan. 2024 – May 2024

Cornell University, Ithaca, NY, USA

### **Introductory Design and Programming for the Web**

*INFO 1300, Prof. Kyle J. Harms and Prof. Benjamin Soltoff*

Aug. 2023 – Dec. 2023

Cornell University, Ithaca, NY, USA

### **Introduction to Data Science**

*INFO 2950, Prof. Benjamin Soltoff*

Jan. 2023 – May 2023

Cornell University, Ithaca, NY, USA

### **Information Ethics, Law, and Policy**

*INFO 1200, Prof. Steven Jackson and Prof. Gili Vidan*

Aug. 2022 – Dec. 2022

Cornell University, Ithaca, NY, USA

### **Computing in Humanities**

*GenEdu1005, Prof. Jason S. Chang*

Feb. 2020 – Jun. 2020

NTU, Taipei, Taiwan

### **Introduction to Game Programming**

*CS550300, Prof. Hung-Kuo Chu*

Sep. 2018 – Jan. 2019

NTHU, Hsinchu, Taiwan

## **PROFESSIONAL SERVICE**

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### **Human-centered AI Workshop**

*Event Planner*

Dec. 2024

Taipei, Taiwan

- Wrote the grant proposal to get funding from the American Institute in Taiwan
- Planned and hosted online mini-sessions with 4 speakers

### **TAICHI X UIST Online Talk**

*Event Planner*

Jul. 2021 – Oct. 2021

Taipei, Taiwan

- Organized a series of online talks regarding UIST (User Interface Software and Technology)
- Invited 13 speakers, including professors, researchers, engineers, and Ph.D. students

### **Communications and Multimedia Laboratory**

*Purchaser*

Jun. 2020 – Jun. 2021

NTU, Taipei, Taiwan

- Researched and acquired equipment and materials worldwide
- Researched and recommended future purchases

### **OpenHCI Workshop**

*Program co-Chair*

Aug. 2019

Taipei, Taiwan

- Organized a five-day workshop that introduced Human-Computer Interaction to select students in Taiwan
- Led a staff team of about 50

## INVITED TALK

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### OpenHCI '23

Aug. 2023

*Invited to share my journey of becoming an international Ph.D student in the USA*

Virtual, Taiwan

- OpenHCI is a student-organized workshop advised by the Taiwanese Association of Computer Human Interaction

### TAICHI '21

Jul. 2021

*Invited to present the two papers accepted by CHI '21 at Best of HCI session*

Virtual, Taiwan

- TAICHI is the top domestic HCI conference in Taiwan

## LANGUAGE

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### Native in Mandarin

**Fluent in English** : IELTS: 7.5; TOEIC: 920; General English Proficiency Test: High-intermediate level